

DUAL-CORE OPTICAL MOTION TESTBED

MECHANICAL PLATFORM & PHYSICAL VERIFICATION

Rotary/translation mechanics, structural integration, physical verification, and demonstrated bench performance

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 7.3.3 | 13 AUGUST 2026

Prepared by	Herder Elektronische Systemen
Document scope	Mechanical platform requirements, physical integration, travel/load verification, and measured bench evidence
Evidence basis	Physical photographs, measured displacement references, load demonstrations, motion results, and engineering review
Distribution	Public portfolio release; CAD, schematics, manufacturing data, and reproduction instructions withheld

DOCUMENT PURPOSE
Defines public functional, performance, interface, safety, verification, and acceptance requirements for the mechanical platform and physical test article while withholding reproduction-level design data.

Functional Requirements and Engineering Context

Public release intent

This FRD preserves the established engineering-report template and provides detailed requirements, rationale, verification evidence, and performance interpretation without publishing CAD, schematics, PCB artwork, dimensions sufficient for reproduction, BOMs, wiring, pinouts, switch settings, assembly procedures, or setup instructions.

Purpose, Scope, and System Boundary

This document governs the public-facing mechanical and physical-verification requirements for the bench-scale optical motion platform. It covers structural integration, rotary/translation functionality, physical travel, representative static loading, visible alignment evidence, and integrated motion results. Electronics are described only at a functional interface level; implementation details remain private.

Boundary element	Included publicly	Withheld
Mechanical structure	Integrated bench platform, functional role, physical verification, measured travel/load evidence	CAD/NX models, manufacturing drawings, dimensions sufficient for reproduction, assembly sequence
Electrical integration	Existence and purpose of motor drive, encoder feedback, controller, and interface electronics	Schematics, PCB layout, grounding implementation, BOM, pinout, wiring, switch settings
Verification	Physical photographs, displacement references, static loading, representative motion response	Calibration/setup procedures, fixtures, internal commissioning instructions
Firmware boundary	System-level requirement that valid feedback and safe actuation states are exchanged with the controller	Source, models, board hooks, cache/MPU/synchronization implementation

1. Platform Role and Delivered Capability

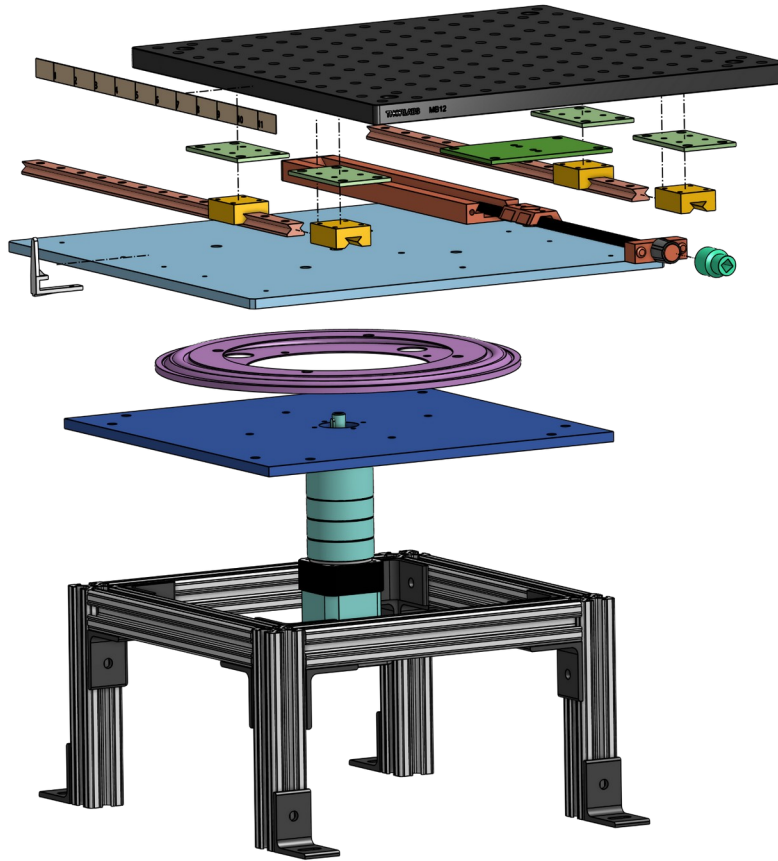


Figure 1. Exploded CAD view of the mechanical testbed assembly.

The exploded view documents subsystem integration and packaging at a portfolio level. Editable CAD and dimensioned manufacturing files are intentionally excluded from the repository.

The platform is a bench-scale rotary/translation test article intended to carry optical or instrumentation payloads while providing controlled angular and linear adjustment. The structural design integrates a rigid lower frame, upper mounting surface, rotary actuation, and a translation mechanism into a single serviceable assembly. Public evidence focuses on what the completed platform does rather than how to reproduce it.

Capability	Public evidence	Engineering interpretation
Integrated rotary/translation platform	Completed assembly photograph and motion records	Both motion functions are physically integrated into one bench-scale article.
Translation travel	Visible reference measurements beyond ± 3 in; detailed record supports approximately $\pm 3 \frac{13}{32}$ in	Demonstrates the intended adjustment range without releasing drive geometry.
Static loading	40 lb offset static-load demonstration	Exceeds the original 25 lb design objective in the demonstrated static condition; not a rated payload specification.
Rotational behavior	Measured integrated-platform response	Confirms motion execution under

		representative test conditions.
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2. Functional and Performance Requirements

ID	Requirement	Rationale	Verification
PL-FR-001	The platform shall provide integrated rotary and translational motion on a rigid bench-scale structure.	The test article must support two-axis experimental positioning without separate fixtures.	Physical inspection and demonstration imagery.
PL-FR-002	The translation mechanism shall demonstrate travel greater than +/-3 inches.	The usable adjustment range must cover the intended bench experiments.	Visible displacement references.
PL-FR-003	The completed prototype shall support the demonstrated 40 lb static offset load without visible loss of structural integrity.	Payload margin is important for representative optics/instrumentation fixtures.	Static-load demonstration.
PL-FR-004	The platform shall retain mechanical alignment sufficiently for repeatable motion testing across the demonstrated travel.	Translation/rotation evidence is only useful if the structure remains mechanically coherent.	Centered/displaced reference photographs and repeated motion tests.
PL-FR-005	The physical interface shall provide encoder feedback and commanded actuation paths compatible with the target controller.	Closed-loop firmware requires a measurable motion state and a controllable actuator interface.	Integrated-system operation; interface review.
PL-FR-006	Public documentation shall not expose reproduction-level mechanical or electrical design information.	Portfolio evidence should prove capability without distributing reusable design IP.	Public-package content audit.

3. Mechanical Integration and Alignment



Figure 2. In-progress mechanical assembly during platform integration.

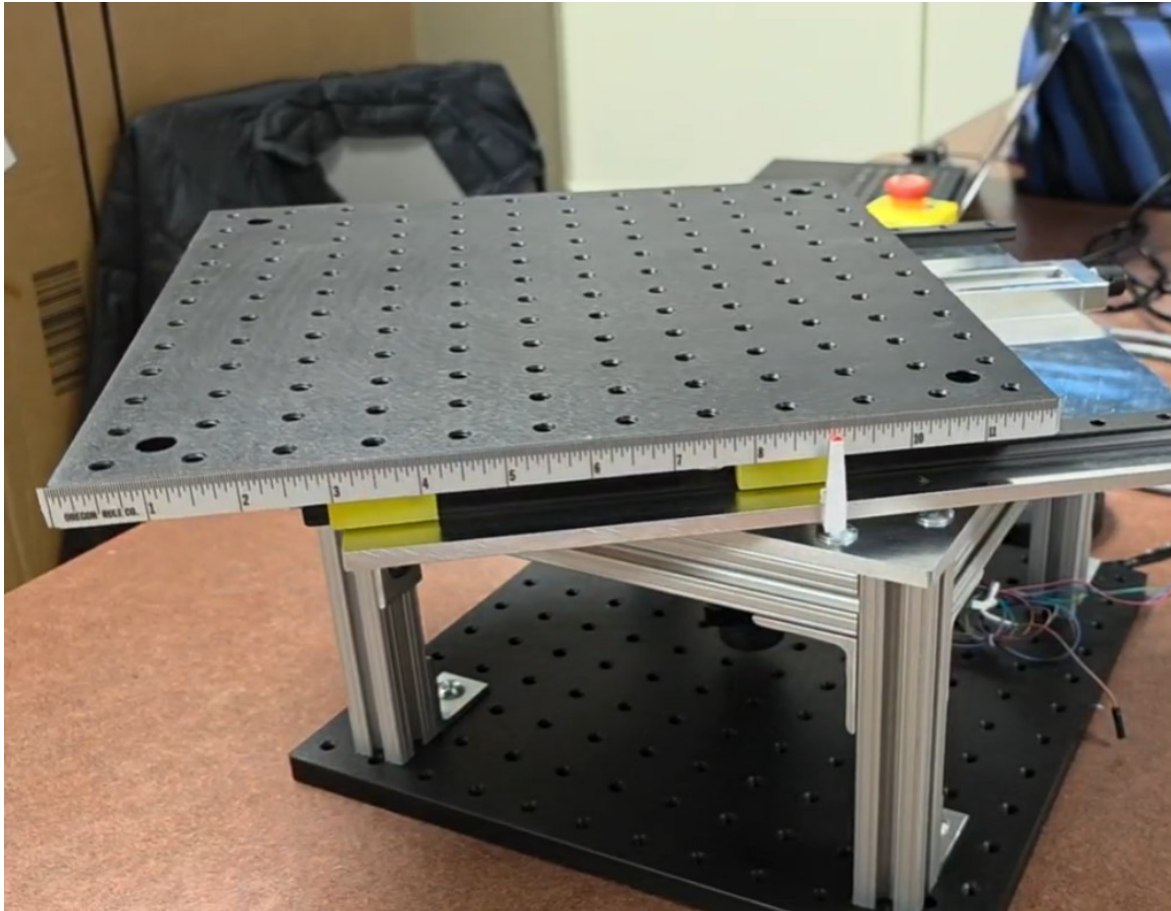


Figure 3. Final assembled optical-motion testbed configuration.

The completed structure uses a rigid frame and upper motion surface to support the moving assembly while maintaining access to the drive and feedback subsystems. The public report intentionally omits the internal CAD stack-up and dimensional design. What matters for verification is that the physical assembly remains coherent across the demonstrated travel and load conditions.

Fabrication evidence is useful because it demonstrates that the architecture progressed beyond CAD or simulation into a physical test article. It is not presented as an assembly guide; fastener selection, alignment procedure, component spacing, and fabrication dimensions remain private.

4. Translation Verification

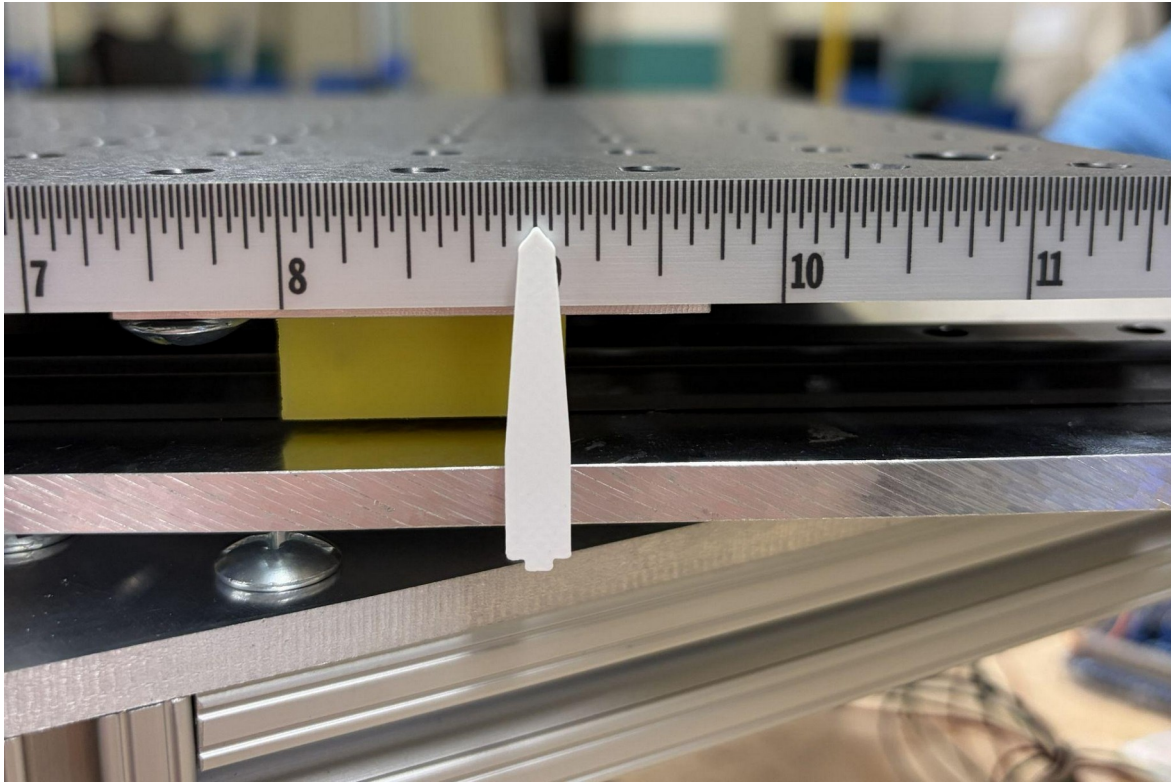


Figure 4. Needle-based translation reference used during displacement and alignment verification.

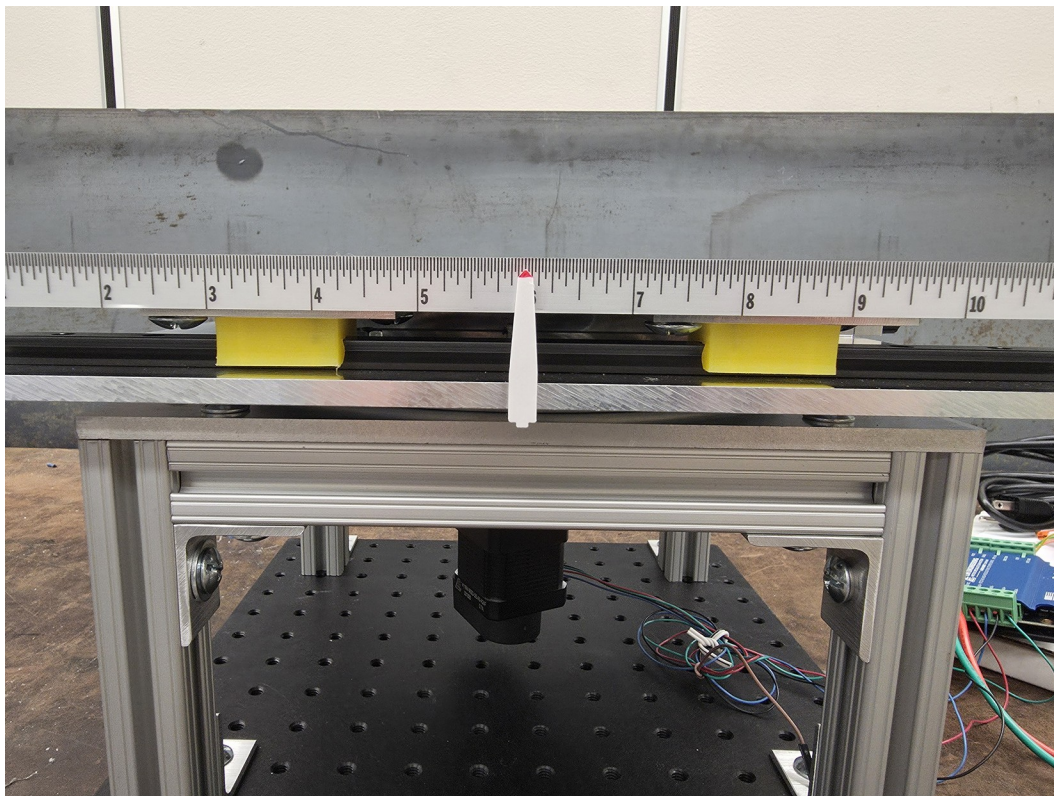


Figure 5. Centered translation reference during physical verification.

Translation was verified using visible reference measurements at multiple stage positions. The retained evidence demonstrates travel beyond ± 3 inches, with detailed project records supporting approximately

+/-3 13/32 inches. The public FRD reports the achieved range but intentionally omits lead-screw geometry, drive ratios, rail spacing, and other reproduction-level dimensions.

Verification aspect	Observed evidence	Boundary of claim
Travel magnitude	More than +/-3 in demonstrated; detailed record approximately +/-3 13/32 in	Prototype travel evidence, not calibrated metrology certification.
Alignment	Reference pointer remains usable across displaced positions	Qualitative/bench alignment evidence.
Repeatability	Repeated motion testing completed on the integrated platform	No formal uncertainty budget is claimed.

5. Static-Load Demonstration



Figure 6. Forty-pound static offset-load demonstration on the completed platform.

The original platform objective included support for a representative payload of 25 lb. The completed prototype was subsequently demonstrated with a 40 lb static offset load. This is a useful margin demonstration, but it is not a rated working-load limit and does not substitute for fatigue, shock, vibration, or long-duration structural qualification.

Load statement	Meaning
25 lb original objective	Design target used to frame representative payload capability.

40 lb demonstrated static load	Observed prototype condition without visible structural failure during the demonstration.
Not claimed	Rated payload, fatigue life, shock/vibration qualification, or safety factor certification.

6. Integrated Motion Evidence

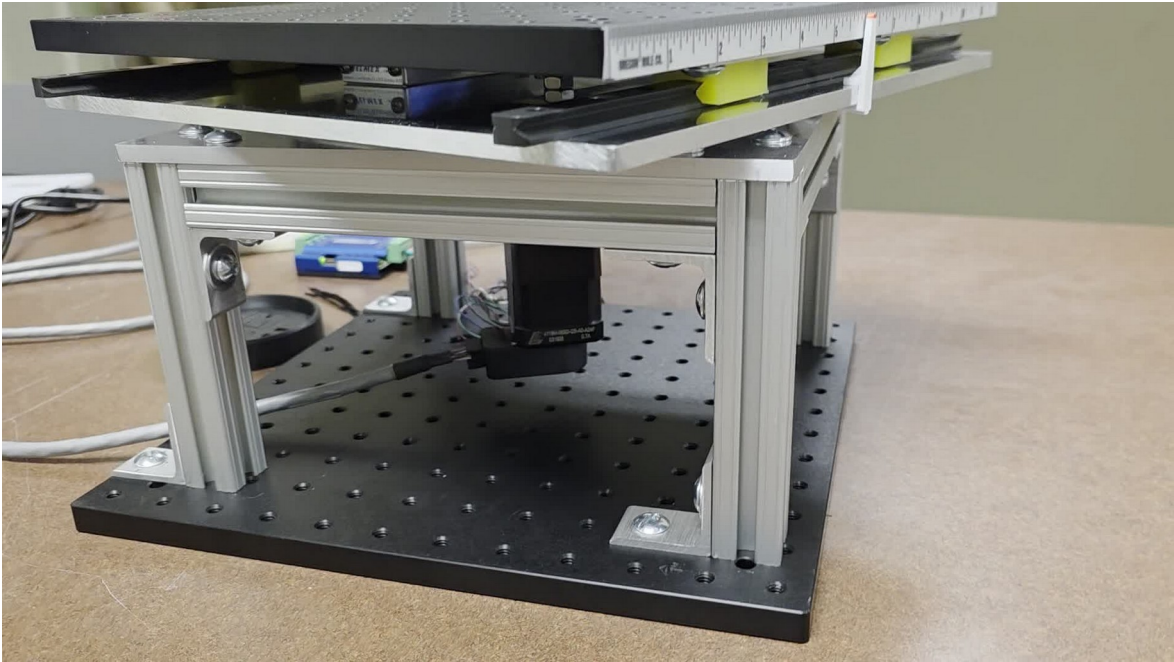
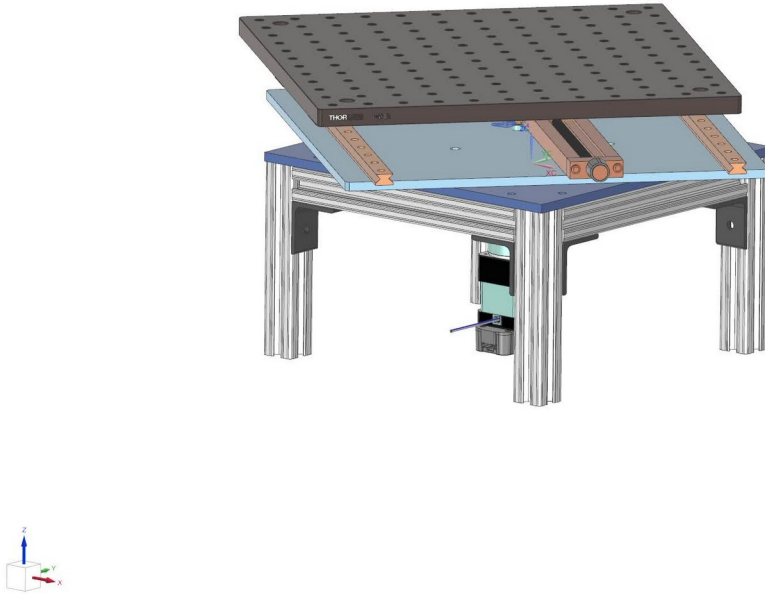


Figure 7. Frame from the integrated system demonstration.

Solution1	
Time	5.880000
Step	588



Solution1 Default Animation

Figure 8. Mechanism-review frame from the NX model animation.

The platform was tested as an integrated motion system rather than only as a static mechanical assembly. Representative response data show commanded angular behavior and the measured system response under physical test conditions. The public package preserves the result while omitting driver configuration, gearing details, electrical wiring, and controller implementation.

The plot should be interpreted as system-level evidence: it demonstrates that the physical platform, feedback path, controller, and actuator were integrated successfully. It is not a component datasheet and does not establish a universal bandwidth or payload/speed envelope.

7. Controller and Feedback Interface Boundary

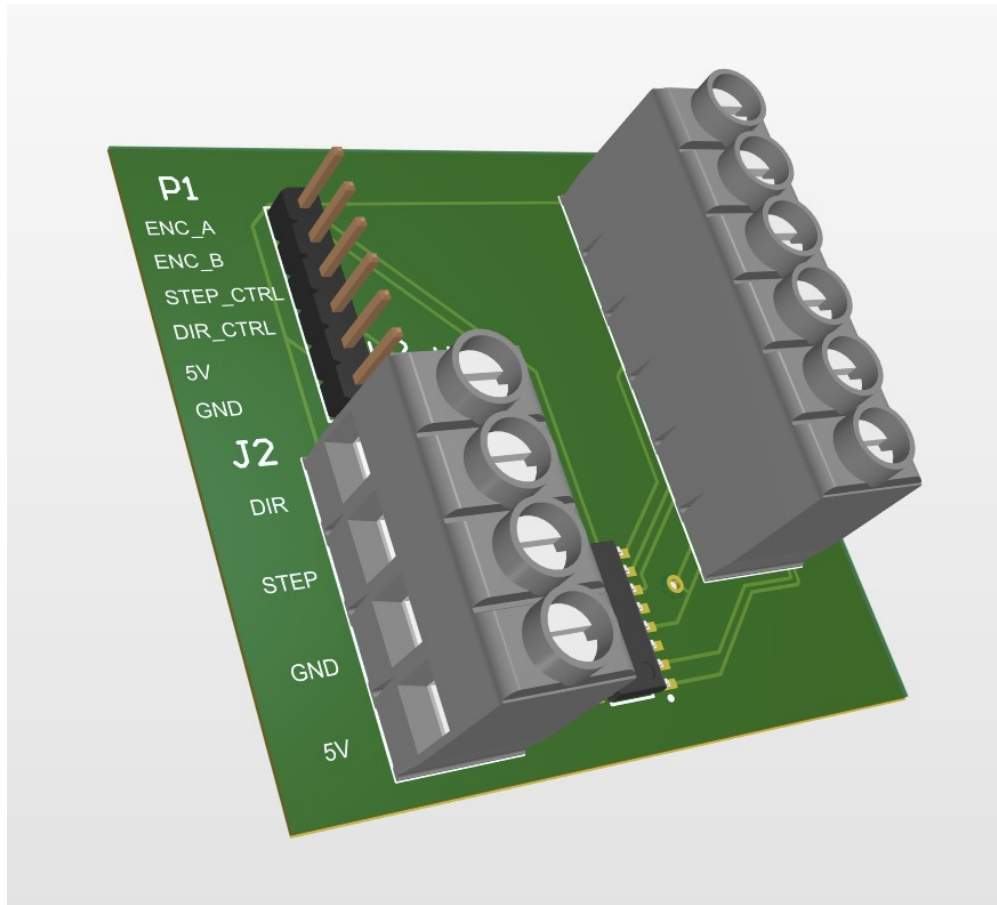


Figure 9. Populated 3D rendering of the custom encoder and STEP/DIR interface PCB.

The PCB is shown as documentary evidence of the delivered interface hardware. Editable EDA files, fabrication outputs, BOMs, pin maps, and reproduction instructions remain outside the public package.

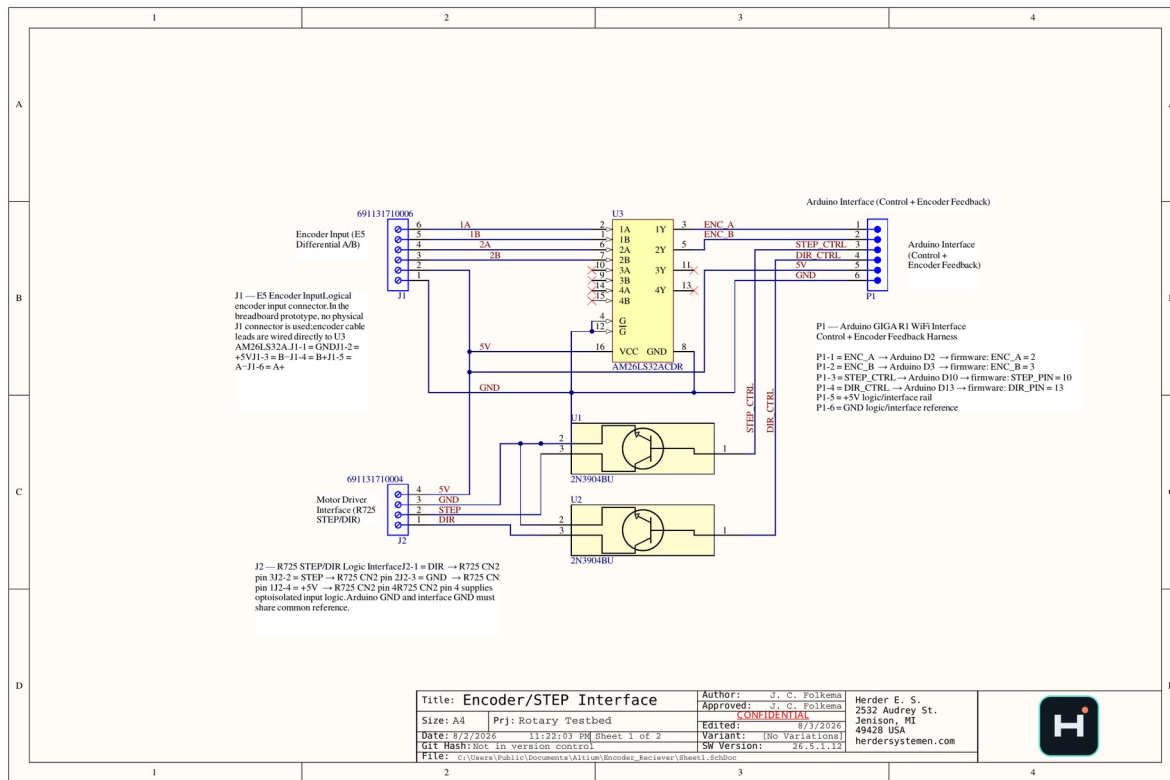


Figure 10. Encoder and STEP/DIR interface schematic.

The schematic is embedded in the FRD for engineering context only. The repository does not distribute the schematic as a standalone source file or provide the associated design database.

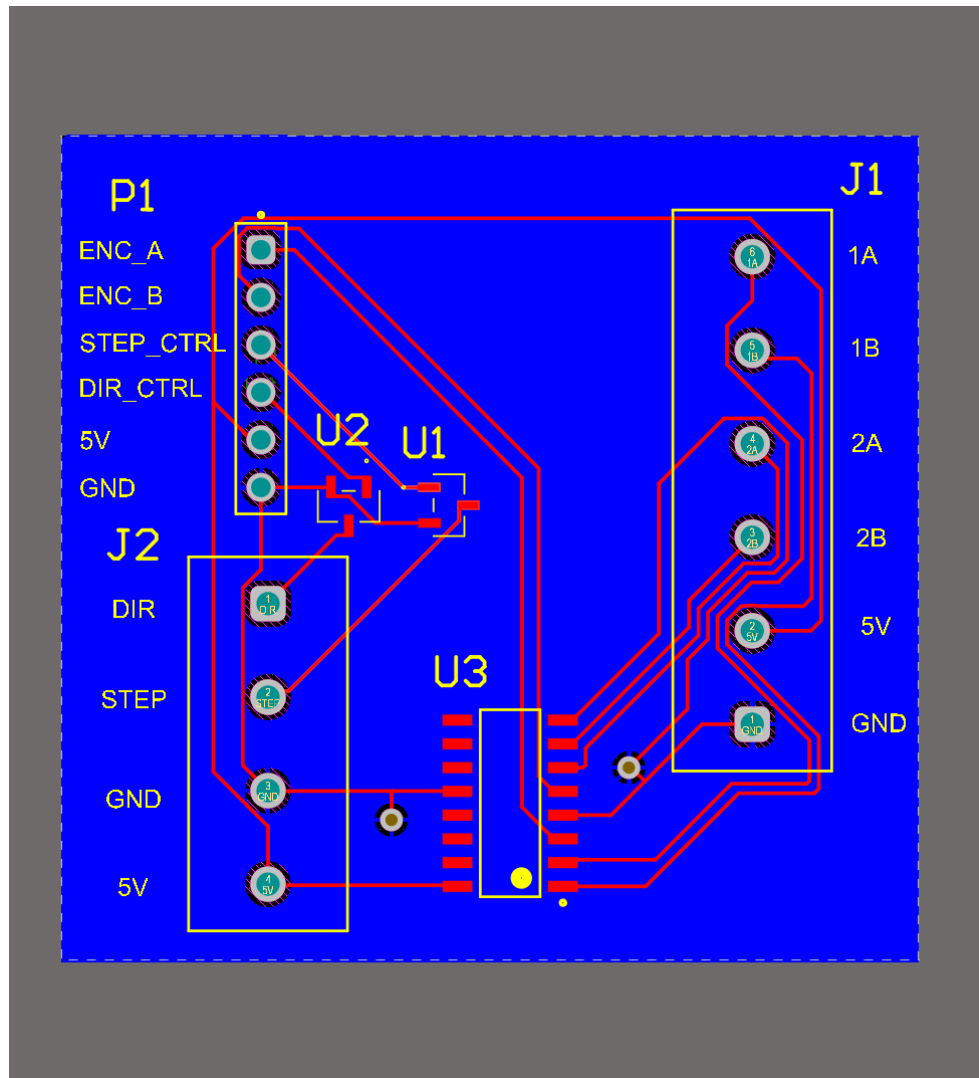


Figure 11. Routed signal layer showing component placement and connector organization.

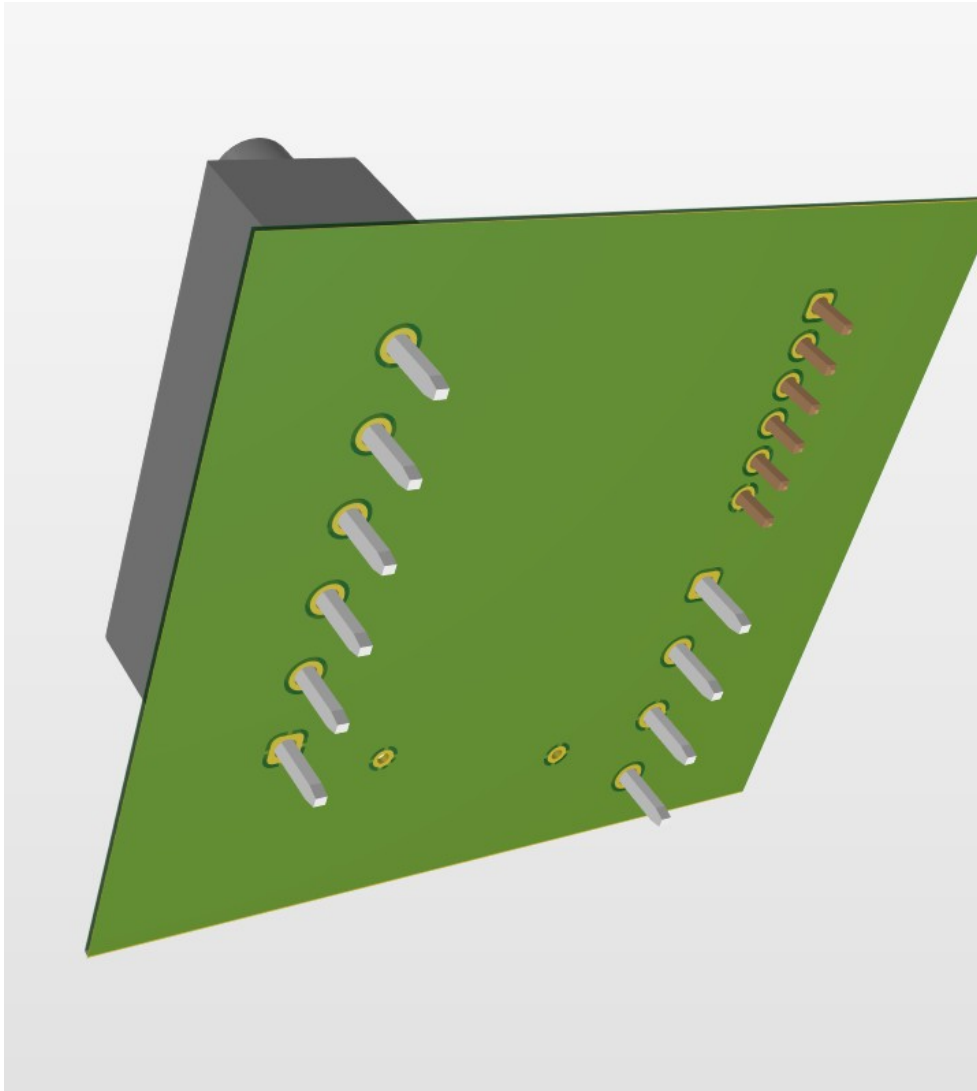


Figure 12. Dedicated solid-copper ground plane on the opposite PCB layer.

The ground-plane figure documents the signal-return strategy while withholding the editable board layout and fabrication data.

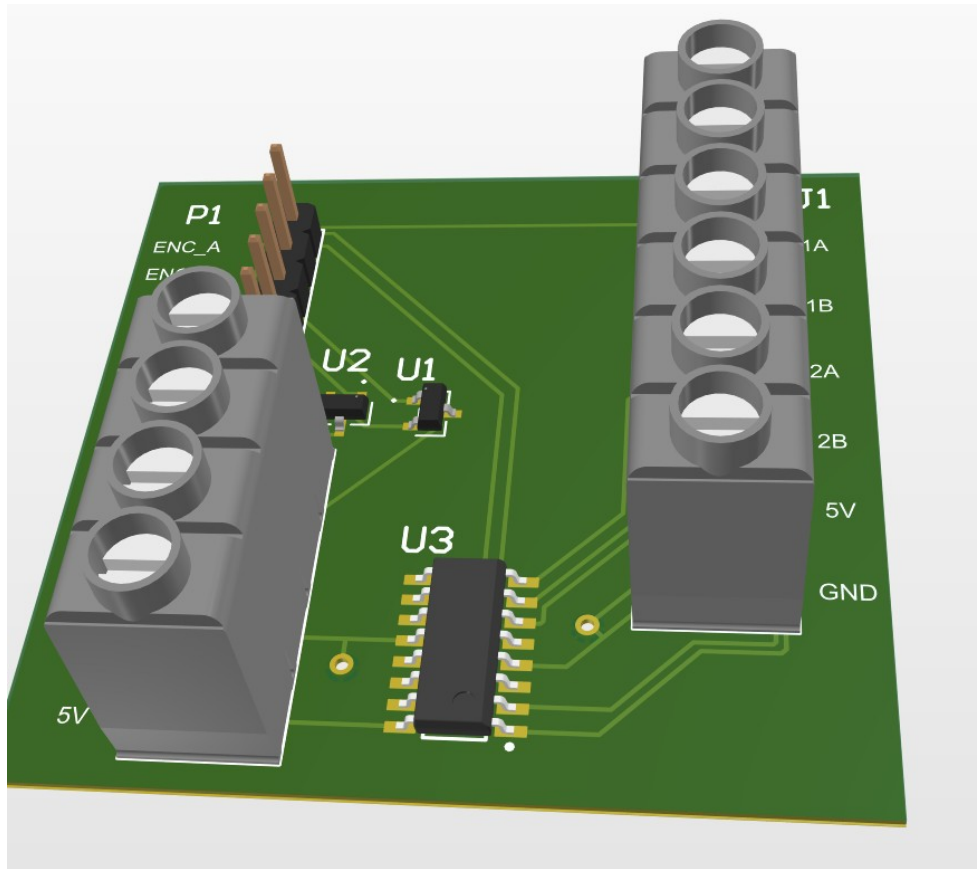


Figure 13. Angled PCB assembly rendering showing the prototype connector and receiver arrangement.

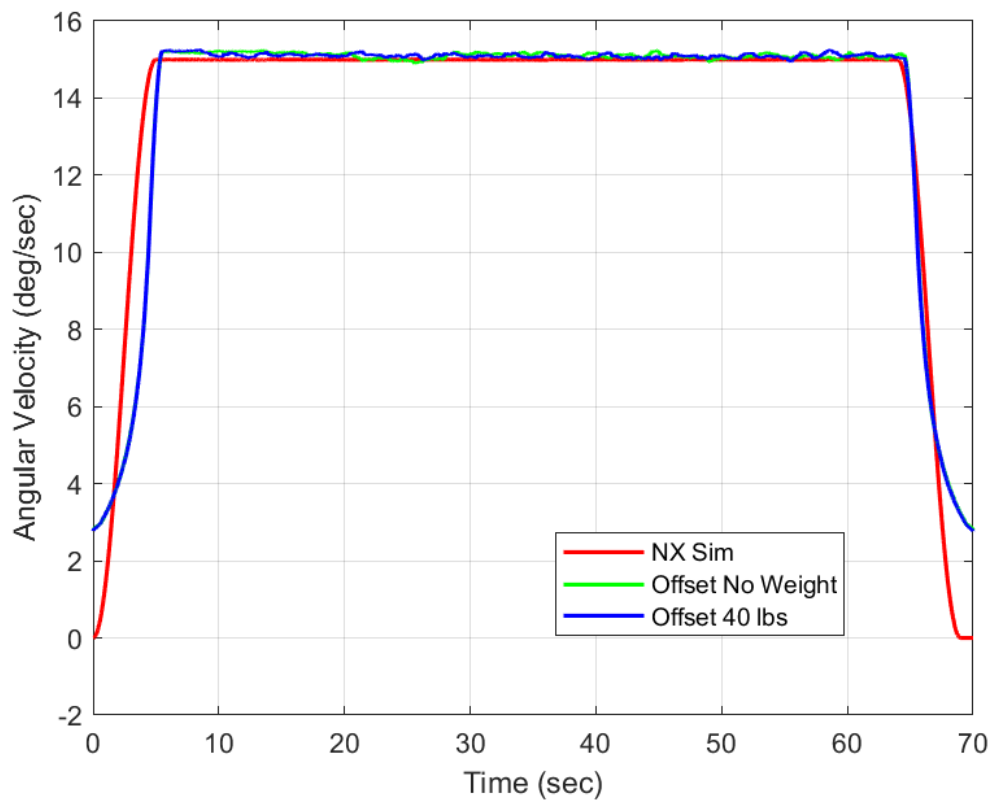


Figure 14. Alternate PCB assembly view.

The mechanical platform depends on two functional interfaces: a commanded actuation path and a measured feedback path. The private controller implementation consumes encoder-derived motion state

and applies state/fault gating before energizing actuation. The public platform FRD intentionally stops at this functional boundary; circuit implementation, connector mapping, grounding strategy, and power-stage details remain private.

Interface responsibility	Platform-side requirement	Firmware-side responsibility
Motion feedback	Provide a physically valid encoder-derived representation of stage motion	Validate freshness/plausibility and use feedback only within permitted control states.
Actuation	Provide a controllable physical drive path capable of executing commanded motion	Validate feedback freshness/plausibility, update the encoder-supervisory control path at its defined cadence, and keep high-rate actuation timing independent of host/service activity.
Fault containment	Physical hardware shall support immediate de-energization of the motion path	Software shall default to disabled/neutral output outside valid running conditions.
Shared data	System shall expose sufficient measured state for test verification	Firmware shall preserve integrity and fault observability across inter-core/host data paths.

8. Verification, Acceptance, and Traceability

Requirement	Evidence	Disposition
PL-FR-001	Completed-platform and fabrication photographs	Demonstrated
PL-FR-002	Multiple translation reference photographs	Demonstrated
PL-FR-003	40 lb static offset-load photograph	Demonstrated, not certified
PL-FR-004	Centered/displaced references and repeated motion evidence	Demonstrated at bench level
PL-FR-005	Integrated closed-loop motion evidence	Demonstrated at system level
PL-FR-006	Public-package audit excluding reproduction-level artifacts	Satisfied by release construction

Acceptance for the public portfolio is evidence-based: the report must show the completed physical article, documented travel, load demonstration, integrated motion behavior, and a clear interface to the companion firmware FRD. It must not include the artifacts needed to rebuild the mechanical/electrical implementation.

9. Limitations and Public Release Controls

- No CAD/NX/Simulink models, schematics, PCB artwork, BOMs, pinouts, DIP-switch tables, wiring diagrams, assembly sequences, dimensions sufficient for reproduction, or setup instructions are included.
- The 40 lb result is a static demonstration, not a rated working-load certification.
- Translation references are bench verification and do not constitute calibrated metrology certification.

- Representative motion results are specific to the tested configuration and are not generalized as universal platform limits.
- Formal shock, vibration, thermal, EMC, fatigue, and production qualification remain outside this portfolio release.

Document Control

Revision	Date	Description
7.3	13 August 2026	Final public platform FRD aligned with the reviewed firmware evidence boundary while preserving mechanical requirements, physical verification, system-interface responsibilities, acceptance criteria, and traceability without reproduction-level design data.